- Porcealin is (clay product) used in line insulators. Transformer bushing pin, switches, Fuse Holder and socket, discs of Electric stove & Kettle and beads are used to insulate heating element. Aluminium is an important refractory material used in furnances.

  Silica glass, Fiber glass, Pyrex are important glass insulation materials used in Capacitor, Radio & TV tubes, Lamps, Laminated boards.

c) Rubber

Rubber is an organic polymer, which elongates on stretching and regains its original shapes after the removal of the stress. Rubber may be classified as natural rubber and synthetic rubber.

d) Mica

Mica is a mineral compound of aluminium silicate with soda potash and magnesium.It has high dielectric strength and low power loss. Mica is used in tapping stator coils, electric irons, hotplates, toasters, motor slot lining and transformer insulation.

e) Asbestos

- It is natural mineral material of fibrous structure and low dielectric.
- It has high dielectric loss and low dielectric strength.
- It is used in electrical machines to withstand temperature, cloth tape, Paper boards, covering on wire of heating element, arcing barrier in switches and breakers.

9.6.2 Liquid Insulating Materials

A) Mineral Insulating Oils

- These oils are obtained from distillation of crude petroleum.It has high oxidation resistance and good thermal stability.
- Transformer oil is used for insulating and cooling transformer (winding and core)
- Cable oil and capacitor oil are other mineral insulating oils.

b) Synthetic Insulating Oils

- Compared to mineral oil, these oils are very cheap and inflammable.
- Askarel is used as coolant in High Voltage Transformer (upto 110° C)
- Aroclors,sovoland sovtol are other synthetic insulating oils.

9.6.3 Gaseous Insulating Materials

A) Air

It is an important insulating material available in nature. It is used in HT lines and capacitor as an insulating material.

b) Nitrogen

- It is chemically inert, prevent oxidation and reduce deterioration.
- It is used in oil filled transformers, capacitors and in cables under pressure.

c) Inert Gases

- They are used in electronic tubes and discharge tubes as insulators

9.7 OPTICAL MATERIALS

Optical materials are becoming increasingly important for communication. In communication, an entire network of optical fibre, LED'S, LASER and detectors

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